

Coding Assignment 2

Due Date: *Monday, September 28 (11.59PM)*

Submission: on Gradescope

Instructions

The coding assignments may be completed in groups of *up to 4 students*. If you do so, please make sure that you include everyone's full name, and that you also *select everyone's name when submitting the assignment on Gradescope*. This ensures that each group member will get a grade assigned and have access to the comments from the graders.

For this assignment, the following items need to be submitted:

1. A **single PDF** that contains your code, all results, plots, and written answers (generated from R Markdown, Jupyter, etc.).
2. The **source file** used to produce the PDF (.Rmd, ipynb, ...).

Problem 1 [50 points]

In this problem, we are going to use a simulated example to illustrate the behavior of *testing* and *training* errors *as the number of predictor **increases***.

- (a) [5 points] Generate a data set with $p = 20$ (independent) covariates, $n = 1,000$ observations and an associated quantitative response vector generated according to the model

$$Y = X\beta + \varepsilon \quad \text{with}$$

$$\beta = (2, -1, 0, 0.8, 0, -1.5, 1.2, 0, 0, 0.3, -0.7, 0, 1.8, 0, 0.5, 0, -1.2, 0, 0.4, 0)$$

The covariates and the ε can be generated using standard normal random variables. (This is similar to what you did in Coding Assignment # 1).

- (b) [5 points] Randomly split the data set into a *training* set containing 200 observations and a *test* set containing 800 observations.

- (c) [10 points] Perform **best subset selection** on the training set, and **plot** the *training* set MSE associated with the best model of each size. Recall that

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- (d) 10 points] **Plot** the *test* set MSE associated with the best model of each size.
- (e) [5 points] For which model size does the test set MSE take on its minimum value? Comment on your results.
- Remark:* If it takes on its minimum value for a model containing only an intercept or a model containing all of the features, then play around with the way that you are generating the data in (a) until you come up with a scenario in which the test set MSE is minimized for an intermediate model size.
- (f) [5 points] How does the model at which the test set MSE is minimized compare to the true model used to generate the data? *Comment on the coefficient values.*
- (g) [10 points] Create a plot displaying $\sqrt{\sum_{j=1}^p (\beta_j - \hat{\beta}_j^r)^2}$ for a range of values of r , where $\hat{\beta}_j^r$ is the j th coefficient estimate for the best model containing r coefficients. Comment on what you observe. How does this compare to the test MSE plot from (d)?

Problem 2 [50 points – 10 points each]

We will use the **Parkinsons Telemonitoring** data set from the UCI Machine Learning Repository:

<https://doi.org/10.24432/C5ZS3N>

The data contain **5,875** biomedical voice measurements from **42** patients with early-stage Parkinson’s disease. The goal is to predict a **textitpatient’s disease severity** (UPDRS score) from a collection of voice measures and a few demographic variables.

You should use the following variables as predictors:

Variable	Description
age	Subject age
sex	Subject gender (0 = male, 1 = female)
test_time	Days since recruitment into the trial
Jitter(%), Jitter(Abs)	Measures of variation
Jitter:RAP, Jitter:PPQ5, Jitter:DDP	in fundamental frequency
Shimmer, Shimmer(dB), Shimmer:APQ3	Measures of variation in
Shimmer:APQ5, Shimmer:APQ11, Shimmer:DDA	amplitude
NHR, HNR	Noise-to-tonal component ratios
RPDE	Nonlinear dynamical complexity measure
DFA	Signal fractal scaling exponent
PPE	Nonlinear measure of fundamental frequency variation

Response: `total_UPDRS` – clinician’s total UPDRS score, linearly interpolated.

Data preparation: Download the data, remove the columns `subject#` and `motor_UPDRS`, and create a random 70 % / 30 % train–test split. Set the random seed to 598.

- Fit a best-subset selection algorithm (up to 8 predictors, excluding the intercept) and report the best model of each size together with its training and test prediction errors (MSE). Keep the output concise.
- Using the models from (a), which model is preferred by (i) AIC, (ii) BIC, (iii) Mallows’ C_p , and (iv) adjusted R^2 ? For each of these four chosen models report both the training and the test MSE.
- Fit a ridge-regression model to predict `total_UPDRS`. Use cross-validation to select the regularization parameter λ .
- Fit a lasso model on the same data. Identify which features are driven to exactly zero.
- Compare the predictive performance (test MSE) of the models selected in (b) with ridge and with lasso. Which approach would you recommend for predicting total UPDRS and why?